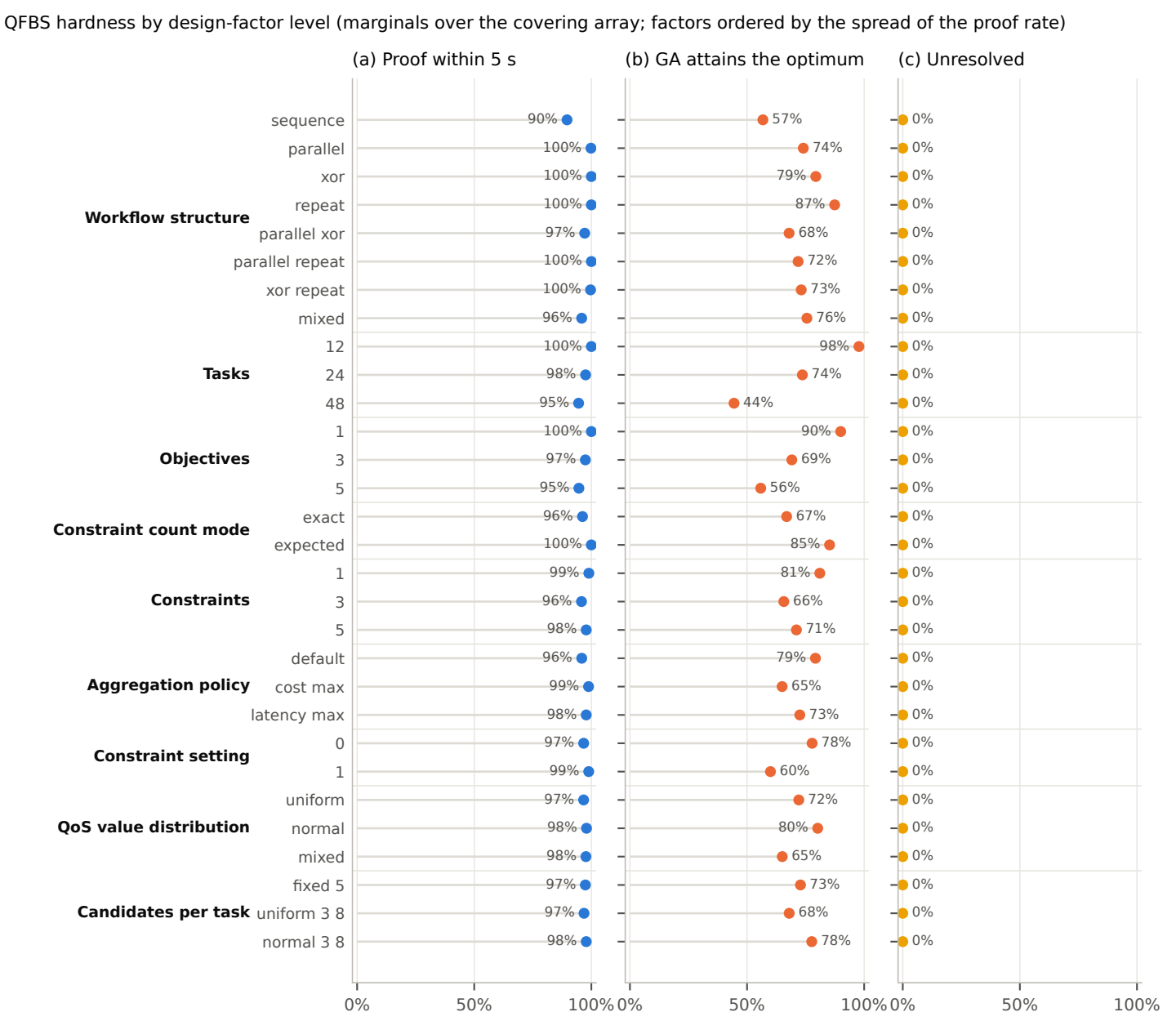
QFBS hardness by design-factor level (marginals over the covering array; factors ordered by the spread of the proof rate)
(a) Proof within 5 s
(b) GA attains the optimum
(c) Unresolved
Workflow structure
sequence 90% 57% 0%
parallel 100% 74% 0%
xor 100% 79% 0%
repeat 100% 87% 0%
parallel xor 97% 68% 0%
parallel repeat 100% 72% 0%
xor repeat 100% 73% 0%
mixed 96% 76% 0%
Tasks
12 100% 98% 0%
24 98% 74% 0%
48 95% 44% 0%
Objectives
1 100% 90% 0%
3 97% 69% 0%
5 95% 56% 0%
Constraint count mode
exact 96% 67% 0%
expected 100% 85% 0%
Constraints
1 99% 81% 0%
3 96% 66% 0%
5 98% 71% 0%
Aggregation policy
default 96% 79% 0%
cost max 99% 65% 0%
latency max 98% 73% 0%
Constraint setting
0 97% 78% 0%
1 99% 60% 0%
QoS value distribution
uniform 97% 72% 0%
normal 98% 80% 0%
mixed 98% 65% 0%
Candidates per task
fixed 5 97% 73% 0%
uniform 3 8 97% 68% 0%
normal 3 8 98% 78% 0%
0%
50%
100%